

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 4 Maths — CA-1 (Medium) — Answer Key

Class : 4

Subject : Maths (Math-Mela)

Max Marks : 30

Duration : 1 Hour

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. A cuboid has 6 faces, 12 edges, and how many corners, following Faces + Corners - Edges = 2? (1 Mark)

- A. 6
B. 8 ✓
C. 10
D. 12

2. Which of these shapes has NO straight sides or angles at all? (1 Mark)

- A. Triangle
B. Square
C. Circle ✓
D. Rectangle

3. If a shape has 6 sides, how many angles does it have? (1 Mark)

- A. 4
B. 5
C. 6 ✓
D. 8

4. If Mini is looking at a building from a helicopter directly above it, which view is she seeing? (1 Mark)

- A. Front view
B. Side view
C. Top view ✓
D. None of these

5. On a grid, an object at row 3, column 1 is located _____. (1 Mark)

- A. in the top-right corner
B. in the third row, first column ✓
C. outside the grid
D. in the first row, third column

6. Which statement about odd and even numbers is TRUE? (1 Mark)

- A. An odd number plus an odd number is always odd
B. An even number plus an even number is always even ✓
C. All numbers are even
D. Odd numbers cannot be added

7. To pay exactly ₹75 using the fewest notes/coins, which combination works best (using 50, 20, 5)? (1 Mark)

- A. 50 + 20 + 5 ✓
B. 20 + 20 + 20 + 5 + 5 + 5
C. 5 × 15
D. 50 + 5 × 5

SECTION B — Short Answer Questions (2 Marks each)

1. Explain how you would identify whether a 2D shape is a pentagon or a hexagon. (0 Marks)

► You count the number of sides (or angles) - a pentagon has 5 sides and 5 angles, while a hexagon has 6 sides and 6 angles, so counting sides helps tell them apart.

2. A cube and a cuboid both have 6 faces, 12 edges, and 8 corners. Explain how you would tell them apart just by looking. (0 Marks)

► A cube has all six faces as equal squares, while a cuboid has rectangular faces that are not all the same size, so checking if all faces look identical helps tell them apart.

3. Explain why the top view of a round object like a ball looks different from the top view of a cube. (0 Marks)

► The top view of a ball (sphere) would show a circle, since it looks round from every angle, while the top view of a cube would show a square, since a cube has flat square faces.

4. If you are standing at the school gate and a friend is on the rooftop looking down, explain why you would see two different views of the same building. (0 Marks)

► You would see the front view (height, doors, windows) from the gate, while your friend would see the top view (the outline/layout of the roof and surroundings) from above, because each position offers a different angle of observation.

5. Explain how a grid can help someone find a hidden object in a "treasure hunt" game. (0 Marks)

► A grid divides an area into labelled rows and columns, so clues like "row 3, column 4" can point to an exact location, making it easier to search systematically rather than randomly.

6. A group of 33 students wants to form pairs for a game. Explain what will happen and why. (0 Marks)

► Since 33 is an odd number, the students cannot form complete pairs - 16 pairs will be formed, and one student will be left without a partner.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. Explain in detail how you would determine the number of faces, edges, and corners of a triangular prism, and describe what makes it different from a triangular pyramid. (0 Marks)

- A triangular prism has 5 faces (2 triangular bases and 3 rectangular side faces), 9 edges, and 6 corners.
- A triangular pyramid has 4 faces (1 triangular base and 3 triangular side faces meeting at a point), 6 edges, and 4 corners.
- The key difference is that a prism has two matching, identical bases connected by rectangular faces, while a pyramid has only one base with triangular faces meeting at a single top point (apex).

2. Describe a real-world scenario where understanding top, front, and side views would be genuinely useful, and explain why. (0 Marks)

- An architect designing a house needs to show top, front, and side views (called elevations) so builders understand the full structure from every angle.
- The top view shows the layout of rooms, the front view shows the height and appearance of the entrance, and the side view shows the depth of the house.
- Without all three views, builders might misunderstand the design, leading to construction mistakes, showing why multiple views are genuinely useful in real life.

SECTION D — Long Answer Questions (5 Marks each)

1. Design a simple 3D model (like a small house or toy) using at least three different 3D shapes learnt in "Shapes Around Us." Describe each shape used, its properties (faces, edges, corners), and why you chose it for that part of your model. (0 Marks)

- For the main body of the house, I would use a cuboid, which has 6 rectangular faces, 12 edges, and 8 corners, providing a stable, box-like structure for walls.
- For the roof, I would use a triangular prism or pyramid shape, which has sloped triangular faces that allow rain and debris to slide off easily.
- For a decorative element like a chimney, I could use a cylinder, which has a curved surface and two flat circular faces, giving a rounded look that contrasts with the cuboid body.
- Each shape was chosen based on its properties - stability from the cuboid, a sloped design from the prism/pyramid, and a rounded contrast from the cylinder - showing how understanding shape properties helps in practical design.